BELASIGNA® 300 Errata

Problem Identifier	Problem Description	Impact	Workaround
Block floating-point support does not work when using the accelerator clock (HEARCLK) prescaler.	When the HEARCLK prescaler is used to rescale the HEARCLK, the block floating-point unit does not rescale results properly when accessing memory.	When using the HEARCLK prescaler, HEAR microcode modules that require block floating-point support produce undefined results. All other HEAR microcode modules work as expected.	Avoid using the HEARCLK prescaler for applications that use microcode requiring block floating-point support. This can be done by load balancing the processing between the CFX and HEAR so that HEARCLK scaling is no longer required. There is no workaround to use HEAR microcode modules that require block floating-point support if the HEARCLK prescaling is enabled.
Chips marked with part number 0W589-001 will have AI4 tied to ground. In these parts, signals to ball E11 will appear on AI3 and AI4 will have no signal.	AI4 is grounded on BelaSigna 300 part number 0W589-001 with AI3 accessible on E11, but on 0W589-002 AI3 is grounded and AI4 is available on E11. The *Hardware Reference Manual for BelaSigna 300* documents that AI3 is grounded because 0W589 chips produced going forward will be revision -002.	BelaSigna 300 software that routes a signal to ball E11 will need to select different signal routing in the input mux setup. No impact on functional behavior is anticipated.	If you are working with OW589-001 chips, set the input mux selector to take the signal from input AI3 rather than AI4. If you are using the BelaSigna 300 ADC autocalibration routine on -001 parts, ground AI3 on E11 to achieve the desired calibration results.
EEPROM block writes using the Program-ROM-based EEPROM library can corrupt data at EEPROM page boundaries if rounding or saturation is enabled. Similarly, 32-bit EEPROM writes using the Program-ROM-based EEPROM library can corrupt data at EEPROM page boundaries if rounding is enabled.	The Program ROM contains an earlier version of the EEPROM library that requires rounding and saturation to be cleared when performing block writes or writing 32-bit words. If a memory word is written to EEPROM as part of a block write or 32-bit write, and the bytes of that word span across an EEPROM page boundary, then certain data words can be corrupted.	When an application writes a word as part of a block write that crosses a page boundary, the bytes written into the second EEPROM page are saturated if saturation is enabled and the first bit in the written word is set. When an application writes a word across a page boundary and rounding is enabled, the last data byte written to the first page is rounded based on the data bytes written to the second EEPROM page.	Use software version 1.2.0 or later of the EEPROM library. Disabling saturation and rounding when writing to EEPROM using the Program ROM version of the EEPROM library block writes will also provide a workaround for this issue.

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When VBAT is set to 1.8V, a high static current (up to 100 μA) is consumed from VBAT when the device is in soft power down mode. The current appears after a few seconds, sometimes showing a peak, and sometimes staying constant.	When the analog circuitry is disabled, VREG and VDBL are tied to VBAT. The ADC comparators are in a comparison state with both inputs at VBAT, and when VBAT is higher than 1.3V, it leads to increased current consumption.	Excessive current is consumed when the device is expected to consume a lower current.	For battery voltages below 1.3V, use the `Sys_SoftPowerdownEnable` macro to enter soft power down mode. For higher battery voltages, use the `Sys_SoftPowerdownEnable_NoAnalog` macro.
Consecutive reads from the math library look up table (LUT) in ROM return repeated data values	When data is read in consecutive cycles from the math library LUT in X memory ROM, the data returned is from the address of the first access.	After the first datum is read from the math library LUT, the same value will be returned for all further consecutive reads from this ROM segment, regardless of address updates.	Ensure there is at least one cycle delay (e.g., insert a NOP instruction) between consecutive reads from the math library LUT in X memory ROM.
HEAR memory conflict is not well managed when accessing data with block floating-point support if the CFX has priority.	When there is a memory conflict on the shared memory bus and the CFX has access priority over the HEAR, HEAR data with block floating-point support might not be correct after the conflict is resolved.	Occasionally, unexpected data might be observed in shared memory.	Structure the data processing in the application to avoid memory conflict when using block floating-point support, or give the HEAR access priority.

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A program does not flow as expected when the interrupt enable bit of the status register (SR) is cleared using the BITCLR instruction in the first delay slot of GOTO.2D, CALL.2D, or RETURN.2D.	When bit 2 of the SR is cleared in the first delay slot of instructions affecting control flow that have two delay slots (GOTO.2D, CALL.2D, and RETURN.2D), the instruction with delay slots does not execute properly. Specifically, the problem appears if the bit is cleared by one of these instructions or macros: • BITCLR SR, 2 • BITCLR SR, SR_IE_POS • Sys_CFX_IntDisable The GOTO.1D, CALL.1D, and LOOP.2D instructions are not affected by this problem. The RETURNI.2D instruction is also unaffected because the BITCLR SR instruction cannot be used in its delay slots since it restores the SR. LOAD and MOVE instructions with SR as the destination are also unaffected.	Control flow is suppressed for control flow instructions with two delay slots (GOTO.2D, CALL.2D, and RETURN.2D) when the first delay slot contains a BITCLR SR, 2 instruction or equivalent. Program control falls through to the instruction after the delay slots, even if the instruction would normally have caused a change in control flow.	Avoid using a BITCLR SR, 2 or equivalent instruction in the first delay slot of GOTO.2D, CALL.2D, and RETURN.2D instructions. Either disable CFX interrupts elsewhere in the program (perhaps by swapping the two delay-slot instructions), or use the interrupt controller's master-enable to disable all interrupts via the digital control register D_INT_ENABLE or D_INT_EBL_0.

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